KAIZEN MATHS

Lesson starter

Complex Numbers

Modulus, Argument and Polar Form

Name _____

S1

Find the modulus and principal argument.

For $z = -7 + 24i$, find $|z|$ and $\arg(z)$ to 3 decimal places.

S2

Write each complex number in polar form.

Write $z = -5 - 12i$ in polar form $r(\cos\theta + i\sin\theta)$, with θ to 3 decimal places.

S3

Write each complex number in Cartesian form.

Write $z = 10(\cos\theta + i\sin\theta)$ in Cartesian form, where θ where $\cos\theta = 3/5$ and $\sin\theta = 4/5$.

S4

Use polar form rules to simplify each expression.

Simplify $3\,\text{cis}(\pi/6) \times 6\,\text{cis}(\pi/6)$.

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ALG-D7A3-L2 · S

Show your working

KAIZEN MATHS

Exit ticket

Complex Numbers

Modulus, Argument and Polar Form

Name

E1

Find the modulus and principal argument.

For z = 8 - 15i, find |z| and arg(z) to 3 decimal places.

E2

Write each complex number in polar form.

Write z = 5 + 12i in polar form r(cos θ + i sin θ), with θ to 3 decimal places.

E3

Write each complex number in Cartesian form.

Write z = 13(cos θ + i sin θ) in Cartesian form, where θ where cos θ = 5/13 and sin θ = 12/13.

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Complex Numbers

Modulus, Argument and Polar Form

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Show your working

Teacher answers & working

Complex Numbers · Modulus, Argument and Polar Form

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S1

Find the modulus and principal argument.

For z = -7 + 24i, find $|z|$ and arg(z) to 3 decimal places.

Answer:

$|z| = 25, \arg(z) = 1.855$ rad

Working:

Use $|z| = \sqrt{(a^2 + b^2)}$.

$|z| = \sqrt{((-7)^2 + 24^2)} = 25$.

Use tan θ = $\frac{24}{-7}$ and choose the correct quadrant for (-7, 24).

The point is in quadrant II.

Principal argument θ = atan2(24, -7) = 1.855 rad.

S2

Write each complex number in polar form.

Write z = -5 - 12i in polar form r(cos θ + i sin θ), with θ to 3 decimal places.

Answer:

$13(\cos -1.966 + i \sin -1.966)$

Working:

Find the modulus first.

$r = |z| = \sqrt{((-5)^2 + (-12)^2)} = 13$.

Find the principal argument using the point (-5, -12) on the Argand diagram.

$\theta = \text{atan2}(-12, -5) = -1.966$ rad.

Therefore z = 13(cos -1.966 + i sin -1.966).

S3

Write each complex number in Cartesian form.

Write z = 10(cos θ + i sin θ) in Cartesian form, where θ where cos θ = 3/5 and sin θ = 4/5.

Answer:

$6 + 8i$

Working:

Use $z = r(\cos \theta + i \sin \theta)$.

Real part = r cos θ = 10 × 3/5 = 6.

Imaginary part = r sin θ = 10 × 4/5 = 8.

Therefore z = 6 + 8i.

Teacher answers & working

Complex Numbers · Modulus, Argument and Polar Form

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S4

Use polar form rules to simplify each expression.

Simplify 3 cis(π/6) × 6 cis(π/6).

Answer:

$18\ \text{cis}(\pi/3)$

Working:

In polar form, multiply moduli and add arguments.

Modulus = 3 × 6 = 18.

Argument = π/6 + π/6 = π/3.

Therefore the result is 18 cis(π/3).

E1

Find the modulus and principal argument.

For z = 8 - 15i, find |z| and arg(z) to 3 decimal places.

Answer:

$|z| = 17,\ \arg(z) = -1.081\ \text{rad}$

Working:

Use $|z| = \sqrt{(a^2 + b^2)}$.

$|z| = \sqrt{(8^2 + (-15)^2)} = 17$.

Use tan θ = $\frac{-15}{8}$ and choose the correct quadrant for (8, -15).

The point is in quadrant IV.

Principal argument θ = atan2(-15, 8) = -1.081 rad.

E2

Write each complex number in polar form.

Write z = 5 + 12i in polar form r(cos θ + i sin θ), with θ to 3 decimal places.

Answer:

$13(\cos 1.176 + i \sin 1.176)$

Working:

Find the modulus first.

$r = |z| = \sqrt{(5^2 + 12^2)} = 13$.

Find the principal argument using the point (5, 12) on the Argand diagram.

$\theta = \text{atan2}(12, 5) = 1.176\ \text{rad}$.

Therefore z = 13(cos 1.176 + i sin 1.176).

Teacher answers & working

Complex Numbers · Modulus, Argument and Polar Form

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E3

Write each complex number in Cartesian form.

Write z = 13(cos θ + i sin θ) in Cartesian form, where θ where cos θ = 5/13 and sin θ = 12/13.

Answer:

$5 + 12i$

Working:

Use $z = r(\cos\theta + i\sin\theta)$.

Real part = r cos θ = 13 × 5/13 = 5.

Imaginary part = r sin θ = 13 × 12/13 = 12.

Therefore z = 5 + 12i.